Does Sports Gambling Reduce Unemployment?

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Abstract

I investigate the legalization of sports gambling over time in the past several years, using an event-study approach to estimate the effect of sports gambling on unemployment from each state. Controlling for macroeconomic factors along with state and time fixed effects, I find that legalization has mixed effects on unemployment, depending on the types of gambling legalized. Additionally, I inspect the impact of legalization on debt and find mixed results there as well, in alignment with current literature.

1 Introduction

Sports betting has grown in popularity over the past several years in the United States and the rest of the globe. Currently, 38 states have legalized sports betting in some capacity, whether retail, online, or both. The overwhelming majority of these states legalized betting within the last decade, signaling a sweeping trend across the nation.

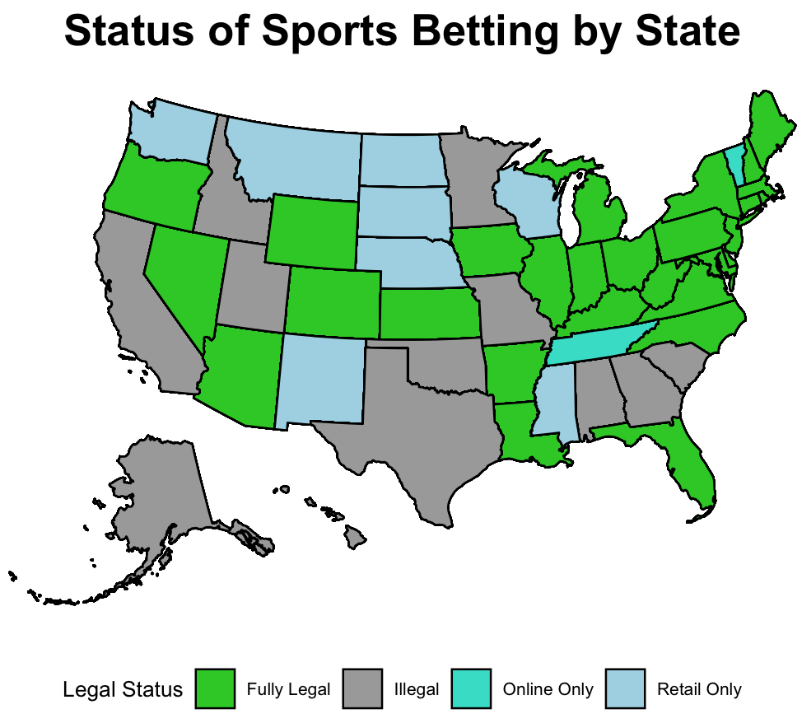

Figure 1: Legal status of sports gambling by state.

Much of the discourse on gambling legalization has focused on its harmful effects, such as damages to personal finances and addiction. The other policy perspective is the revenues that governments would be able to gather from tax revenues, licenses, and other fees. Few studies have been conducted on the impacts of state legalization, to the point where I can only refer to a handful of papers that analyzed the issue using different (yet similar) perspectives. There are even fewer resources that look at the employment effects of gambling legalization.

1.1 The Rationale of Gambling for Employment

Within the gambling industry, the majority of jobs are generated through the creation and management of physical gambling locations such as casinos or booths, so bookkeepers can manage bets while a sporting event transpires. In 2023, nearly 600 of the 700,000 jobs within the gambling industry fit this description (Oxford Economics 2023). In modern times, the scope of work in the gambling industry has grown, and many gambling operators are now strictly online in their work.

Another argument for legalization of gambling is that there are *indirect* employment benefits that help the local economy. The American Gaming Association (AGA) claims that about 1.8 million non-gambling industry jobs are affected by the legalization and existence of the gambling industry. The AGA also claims $178 billion in indirect spending compared to the $150 billion directly(Oxford Economics 2023). If true, this would make the legalization of gambling very attractive to local policymakers, as it may bring economic growth to other industries in the region. This may also have some added effect on employment.

1.2 Literature Review

Although there has yet to be a paper that examines the employment effects of sports betting legalization (especially at the state level), there are some papers that are similar to help construct a working hypothesis and theory behind gambling legalization.

The theory that gambling legalization contributes to employment may not be as straightforward as the AGA would like to put it. The macroeconomic danger of gambling legalization is what spending and employment it could take away from other industries. A paper from the Donahue Institute from the University of Massachusetts found that when the state of Massachusetts legalized sports betting, there were only 118 net new jobs. This was due to a loss of 2,465 non-gambling jobs due to reduced spending in those industries, offset by 1,861 jobs from reinvested gambling tax revenues and the 722 direct hires in the gambling industry. Gambling could be seen as a substitute for other forms of entertainment on which an individual might spend their money.[1] Consequently, spending could go from one industry to another, bringing jobs from one industry to another without necessarily impacting net employment figures.

Aside from employment, there are other economic studies of gambling legalization on consumers. Hollenbeck, Larsen, and Proserpio (2024) looked at the impact of legalization on consumer financial health (credit scores, debt, etc.) through looking at a dataset of consumer finances of roughly seven million bettors. They found a small negative effect to credit scores by 0.3% that is significant. The authors also find that credit limits also decreased. In order to pursue more specific results, the study splits into analyzing two main groups by looking at legalization (retail and online) as well as the incorporation of online betting options. Doing this, the authors found that the dip in credit scores was tripled due to exposure to online betting. In terms of debt, they found that auto-loans and credit-card debt suffered, where there was a higher likelihood of delinquency (falling behind on payments). When considering total legalization (retail and online),

[1] This is somewhat contentious due to the nature of agglomerated industries. Las Vegas and Atlantic City are exceptions to this theory, as their tourism, restaurant, and entertainment industries are reliant on the gambling scene. However, it can be argued that since these cities prioritize gambling over other activities (something no other major city engages in), this is an exception rather than a counterexample.

they only found that auto-loans were more delinquent. Finally, the authors also found similar results for the likelihood of bankruptcy.

R. Baker et al. (2024) had a more broad study, looking at not only personal finances, but also investing and consumption behaviors in industries related to sports betting. The authors were able to look at investing patterns in cryptocurrency, engagement with poker (what we may consider as traditional gambling), as well as consumption of restaurant and entertainment services. What the authors found aligned with the findings that Hollenbeck, Larsen, and Proserpio (2024) had. Most importantly, they found a significant increase in the consumption of restaurant, entertainment, and cable/telecommunication services, which is an indication that there is a potentially beneficial effect to local economies because of sports gambling. This gives the argument made by the AGA some merit, as there is a significant increase in indirect spending due to gambling, but possibly not at the level that the AGA believes (Oxford Economics 2023).

Both papers implement a staggered event study design in order to make best use of each state legalizing gambling. This was not the only similarity. In particular, they both implemented state and time fixed effects in order to account for the possibility that treatment selection was not random. One possible reason for why states may legalize sports betting is because they are enticed by the growth in tax revenues, which would be attractive in years when the state government fails to garner enough revenue. This is one of many checks that are implemented in their papers to ensure a valid implementation of their methodology.

1.2.1 Research Question

The aim of this paper is to investigate whether the legalization of sports gambling has had an effect on unemployment. As mentioned before, R. Baker et al. (2024) found that was a significant increase in consumption of restaurant and entertainment services, which could provide enough stimulus to the local economy to provide jobs. However, there could also be substitution effects that move employment from one industry to another. Another theory is that gambling causes people to look for work, as they need to fuel their addiction and pay off their debts. Hence, it is better to reduce the scope of our question to simply: Does legalizing sports betting bring jobs to individuals who *need* employment? In other words, we shift our focus to the individuals who are looking for work rather than those who are moving from one industry to another. Inspecting unemployment rates can help reveal the reality behind the apparent job growth that the AGA claims.

2 Methodology

2.1 Data

The data for the treatment was assembled by myself based on legalization years/status from the American Gambling Association as well as the two papers mentioned before (Hollenbeck, Larsen, and Proserpio 2024; R. Baker et al. 2024). For the controls and unemployment, all data was retrieved using the FRED API developed by the St. Louis Fed. I also conduct supplemental analysis using per-capita debt levels by state from the Fed New York's Micro data Center.

In terms of time, I only consider the years of legalization since 2016 to

2023, looking at data *yearly* as opposed to quarterly for the other studies.[2] Due to the availability of the control data (in particular education and population), I am forced to run the models using years instead of quarters.

2.1.1 Effective Tax Rates on Gambling Revenues

I also possess data on the effective tax rates on some states since 2018. However, this data is not complete, as certain states are missing data, or do not charge a direct tax on gambling revenue, instead delegating the collection through income taxes or some other means. This makes the utilization of effective tax rates to be difficult, as while it does improve upon just using the nominal tax rate proposed by each state's legislature, it still could miss some nuances of tax collection.

Regardless, I utilize a dataset from Legal Sports Rulebook, where they collected the handles, holds, taxes and revenues from as many states they could from 2018 onward. This means that the number of observations that could be utilized decreases significantly to only 96. Consequently, I simply test for an association between the tax rate and unemployment rate with state and time fixed effects.

2.1.2 Exclusion of Nevada

In my data, I choose to exclude Nevada from the study due to its history with gambling. Due to Nevada having legalized retail (traditional) gambling in 1949 and the legalization of sports gambling being in 2010, it is far too late in terms of the consideration of the time frame I am looking at. This is not out of the ordinary, as Hollenbeck, Larsen, and Proserpio (2024) also excludes Nevada for being early in terms of legalization.

[2]Data for education (bachelor's degree or higher) stopped being collected in 2023. No other proxy could be found that was more recent.

The economic structure of Nevada is quite different from other states. Unlike gambling being a supplemental industry that is introduced in order to spur economic activity, gambling is the core focus of Nevada's economy, which makes its influence on the unemployment rate extremely strong. Because of this, it would disrupt the generality that this analysis aims to achieve. Additionally, Nevada's unemployment is higher than average, which can reduce the confidence in results when applying our models (see Figures 10 and 11). This serves as my rationale to exclude it from this study.

2.2 Identification Strategy

Similar to the two papers that I have mentioned above, I will be implementing a staggered event study using the estimators outlined by Sun and Abraham (2021), which is a regression-based solution towards the issues of heterogeneous treatment effects in a differences-in-differences study. This type of method is necessary due to the variability of "treatment" or legalization for each state.[3]

[3]Additionally, I do actually run the models using regular two-way fixed effects (TWFE) in order to show the biased results, the course of the bias was due to an inappropriate application of the model.

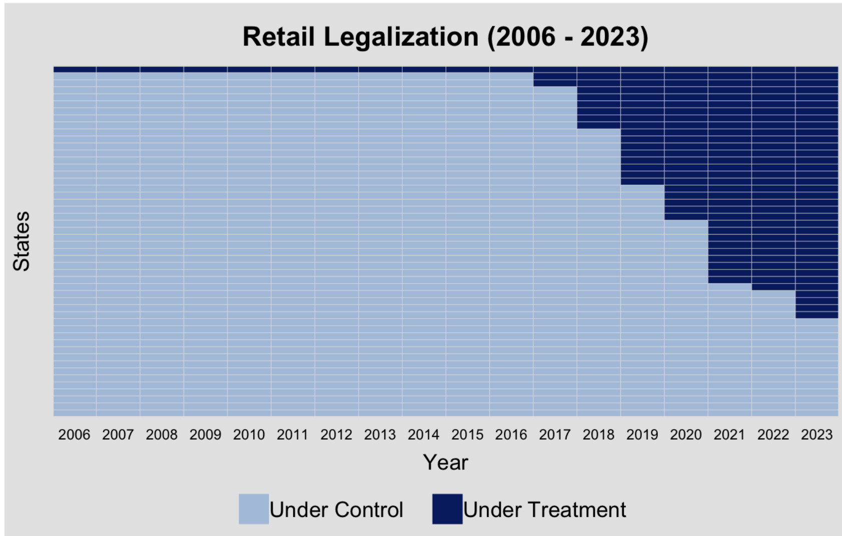

Figure 2: General pattern of retail sports betting legalization over time, acting as a staggered treatment.

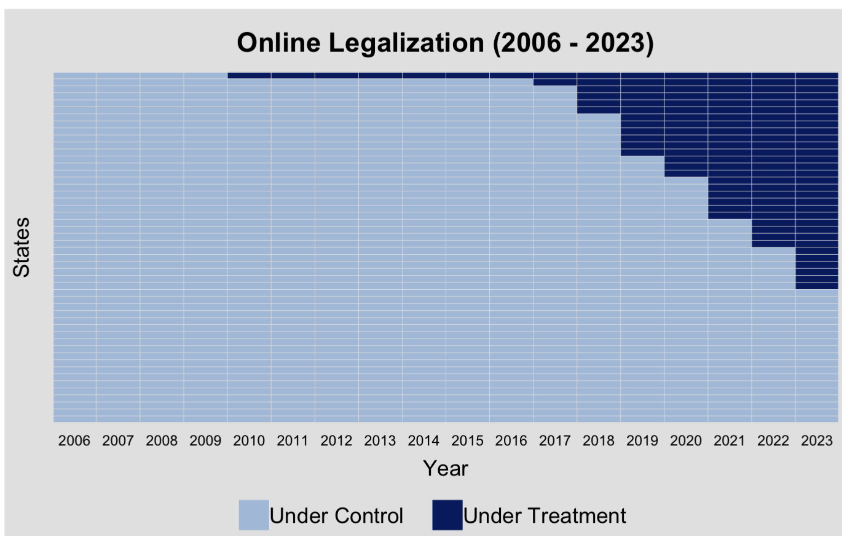

Figure 3: General pattern of online sports betting legalization over time, acting as a staggered treatment.

I approach my analysis with a similar approach that Hollenbeck, Larsen, and Proserpio (2024) took by splitting my analysis into retail and online legalization. This way, I am able to have a more specific analysis of the effects on unemployment. I also include controls for general macroeconomic indicators such as real GDP (chained 2017 $USD), population, and the proportion of the population with a bachelor's degree or higher (a proxy measure for education). In certain models, when there was not an issue with multicollinearity, I factor other variables as well. In all models, I use state and time fixed effects with clustered standard errors using the states.

2.2.1 Brief Overview of Sun and Abraham (2021)

Sun and Abraham (2021) address the issue of time-heterogeneous treatment effects in a differences-in-differences model using the interactions of the treatment and the years (whether that be pre-treatment or post-treatment) involved in the model. This way, the years of when a state has not yet legalized gambling can be used as a control (in the pre-treatment years) while also being included in the post-treatment interactions. A small benefit from these models is that we are able to observe each pre-trend year in order to see if the pre-trends are parallel (at the very least). Of course, this does not relax the assumption of the counterfactual trends still being parallel.

Additionally, another benefit of Sun and Abraham's method is how they deal with other heterogeneous treatment effects using cohorts. In my model, the cohorts are the groups of observations that were treated at the same time. Thus, when we estimate the differences-in-differences, we have time and cohort level interactions that can be used to estimate the cohort average treatment effect on the treated (CATT). Using the CATT, we can use a weighted average in order to calculate the ATT, which is what we would usually get from the interaction of the treatment and period in a canonical TWFE regression. This results in a robust estimation of the treatment effect.[4]

2.2.2 A Note on Assumptions and Heterogeneous Treatment Effects

Other than parallel trends, there are other assumptions we must relax. One of them is anticipation effects (which, if present, can throw off the pre-treatment and post-treatment differences). Another is for the treatment to be exogenous, meaning that there is no spillover effects into the other states. Given the nature of gambling, it could very well be possible that legalization can impact the economies of other states. In terms of cross-checks and other tests in order to ensure the validity of the results, Hollenbeck, Larsen, and Proserpio (2024) used COVID-19 assistance programs (in particular unemployment insurance) as a test to see if there was any ”timing” in legalization. The authors then estimated the ATT for gambling using the method developed by Callaway and Sant’Anna (2021), which is a nonparametric method towards resolving the heterogeneous treatment effects of the usual differences in differences study.

Another potential issue not addressed in this model is the inclusion of more variables to account for differing tax rates and regulations. In both papers I have mentioned previously, the authors were able to do heterogeneity analysis using the consumer demographics (low savings, race, age, gender, etc.), making the estimated treatment effect more specific. Due to the limitations of my data, I cannot do this. However, I could have implemented a third difference in tax rates. This is discussed more in the conclusion.

[4] This method is extremely ”black box,” due to the weighting scheme being algorithmic in nature.

3 Results

Using the model outlined by Sun and Abraham, I find little to no evidence that the legalization of sports betting has an effect on unemployment rates (Table 2). For the model on retail gambling legalization, I find that the parallel trends assumption is shaky, particularly in the pre-trends of the retail model (see Figure 4). However, the standard errors are large enough to make it valid. The noise may be explained by the types of retail gambling there are, especially with certain states only allowing Native American tribes to own gambling licenses, granting them monopoly privileges. Because of this, the effect on unemployment would be harder to discern. This is one of the many heterogeneous treatment effects not necessarily captured in the model.

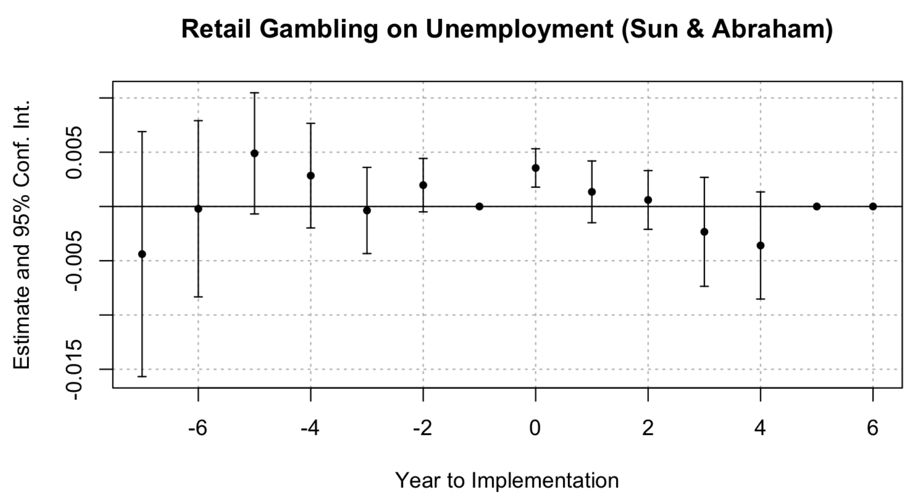

Figure 4: Retail Gambling Legalization on Unemployment

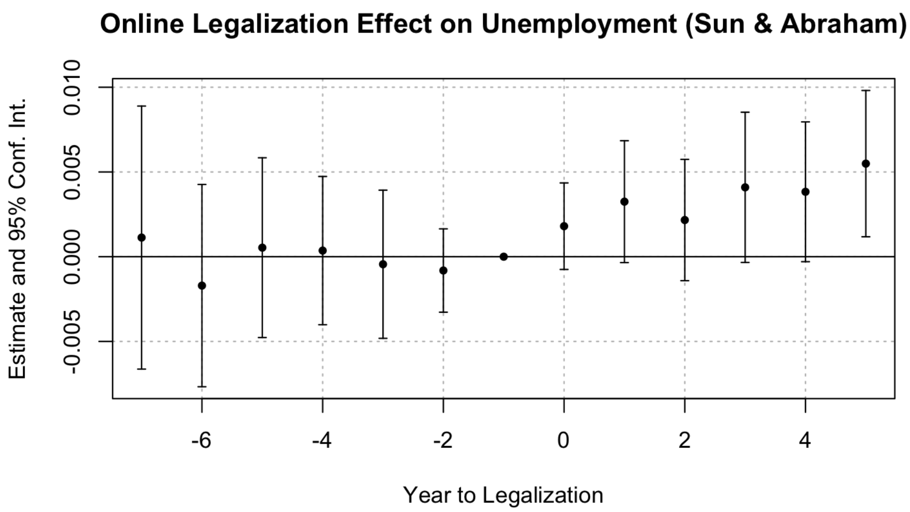

Figure 5: Online Gambling Legalization on Unemployment

The online gambling model is somewhat similar, with the pre-trends of the model now being more stable. I find that after treatment, there are some years that indicate that there was a small increase in unemployment, around 0.3% to 0.5% varying by post-treatment year. When I estimate the average treatment effect on the treated (ATT) using the weighted average of the CATT, it shows that the effect on unemployment is small, but significant (a 0.3%) increase in unemployment as a result of online gambling being legalized. Within the theoretical framework discussed earlier, what likely drives this result is spending leaving the local economy, impacting jobs within the state. When retail betting locations are opened, there could be a substitution effect that takes place, resulting in local spending not changing that much. This would explain the lack of a significant change in Figure 4. Continuing this theory, online sports betting could similarly introduce a substitution effect, but *not* have that spending stay within the local economy. As a result, state unemployment increases in the long-run as the nature of consumer spending evolves to rely less on goods/services in the state.

3.1 Supplemental Analysis for Robustness

Additionally, I test the unemployment and employment level. The first is a reformulation of the original purpose of the paper as an additional check for robustness. Estimating the model's effect on employment is done to see if there was any *net gain or loss* in jobs, as the reports done by the AGA. When I inspect changes in the employment level, there is a brief period shortly after the year of implementation where employment actually decreases.

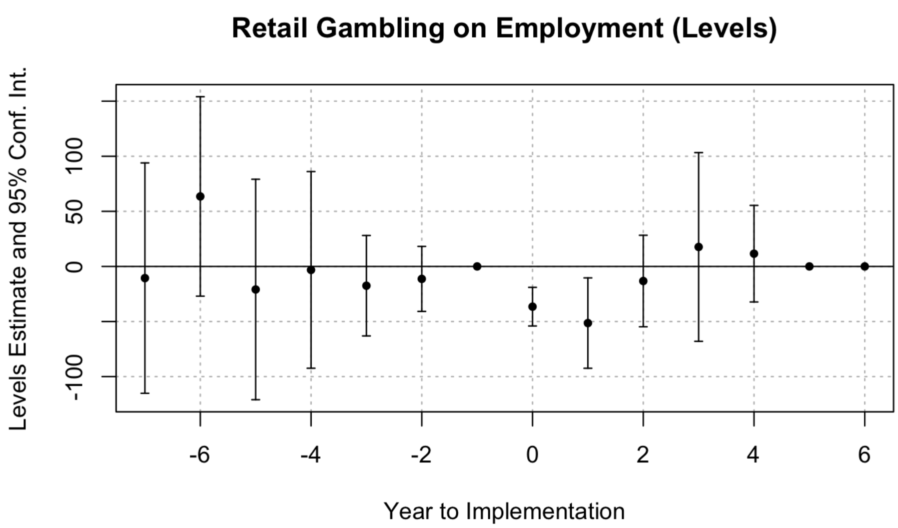

Figure 6: Retail Gambling Legalization on the Employment Level

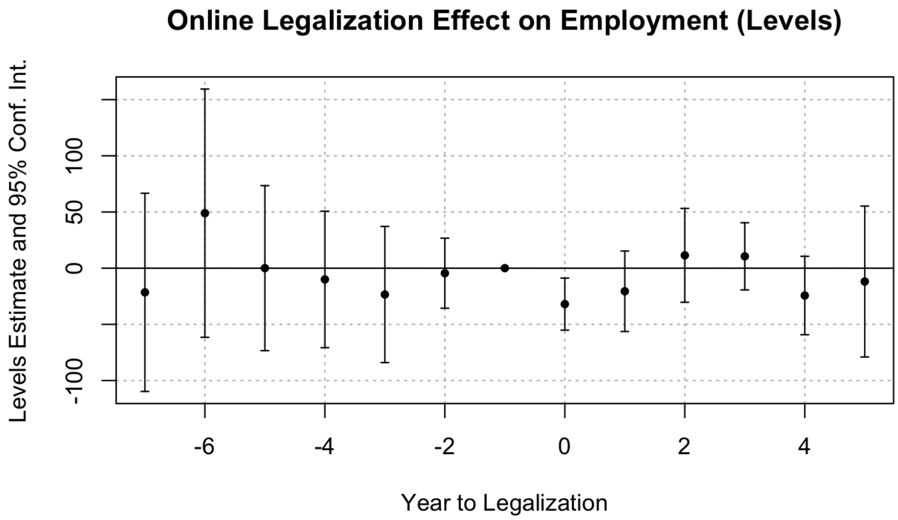

Figure 7: Online Gambling Legalization on the Employment Level

This effect is not carried on in the subsequent years and fizzles out (Table 3). For the unemployment level in retail, I observe no significant effect other than the year of implementation (Table 4).

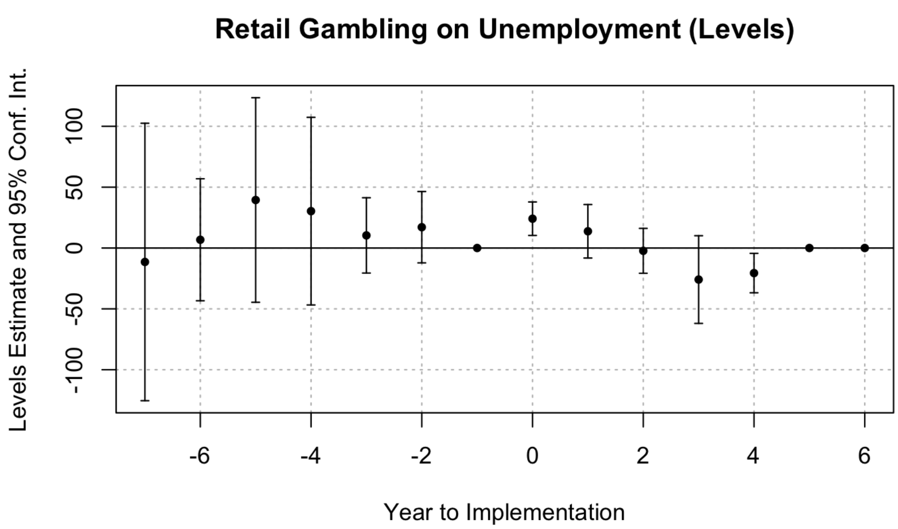

Figure 8: Retail Gambling Legalization on the Unemployment Level

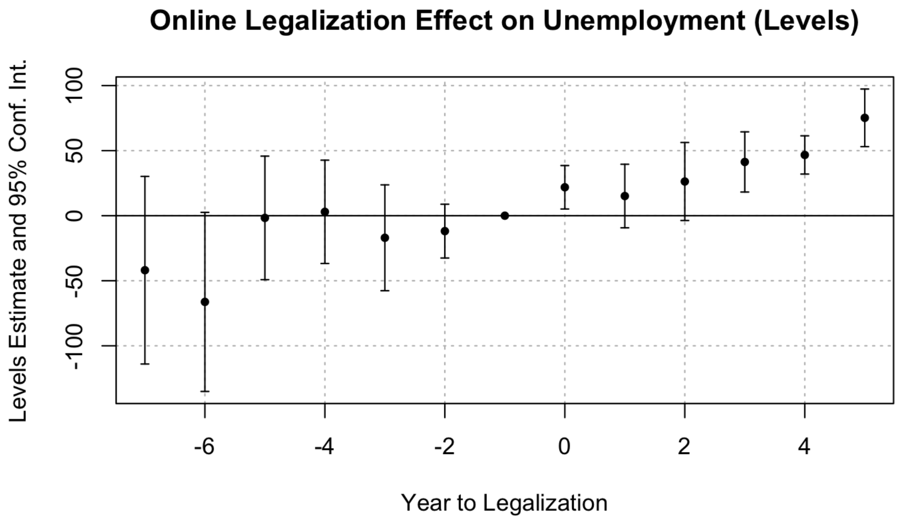

Figure 9: Online Gambling Legalization on the Unemployment Level

This supports my primary analysis using the unemployment rate. In general, the results of the unemployment level mirror the unemployment rate, with the online legalization having an effect.

These results support the substitution theory. The findings coincide with the literature from the Donahue Institute, which found a substitution effect when sports betting was legalized in Massachusetts (Donahue Institute 2025). There, they found that although there was a *net gain* in jobs in the state, there were considerable losses that were recuperated from jobs created from government spending. Given the scope of this paper's analysis, it's impossible to make the distinction between gambling related employment nor jobs created by the government. However, the lack of any effect on aggregate employment is an indication that there is likely some sort of substitution effect going on, with the net gain/loss dependent on other state policies. The brief period of less employment could be explained from the institutional lag it takes for government projects to materialize into creating jobs.

However, the rise in unemployment for online sports betting does not follow the substitution theory. This could provide evidence for the other theory: gambling enables people to find work in order to further gamble or pay off debts. Perhaps, since there is an ample amount of jobs provided by retail betting, the effect on unemployment is null. For online sports betting, there could be a movement in spending outside the local economy, which causes unemployment to rise, but no jobs created within the state to meet the demand. There is also the possibility of the two types of sports betting interacting with each other. This is discussed in the conclusion.

Finally, I use the effective interest rates to determine if they have any relationship with the unemployment rate. I define a basic OLS model with time and state fixed effects to see if there is any significant relationship between the effective tax rate levied on gambling vendors and the unemployment rate (see Appendix). With the model, I find no correlation whatsoever with the unemployment rate, an indication (along with the fixed effects) that there is likely no causal relationship with the unemployment rate at all (Table 5). However, there could be a case for a causal relationship. If a state is able to find the tax rate to maximize tax revenues (dependent on the elasticity of sports betting), then they would be able to leverage those earnings to other productive activities like job training programs and public works projects. With our model, I can only conclude that higher or lower effective tax rates do not influence the unemployment rate in any way.

3.2 Evaluating Previous Findings by Hollenbeck, Larsen, and Proserpio (2024) and R. Baker et al. (2024)

In addition to the models on unemployment, I also inspect legalization's impact on credit card debt per capita. In general, I see that there is not much of an effect on credit card debt, especially for retail gambling (see Figure 12). For that model in particular, the pre-trends were slightly nonparallel. The model also had the resulting effect when the ATT is calculated, which finds that credit card debt per capita actually *decreases* by around $44 for retail gambling. This is likely because banks and other lenders restrict their credit in order to prevent delinquencies and default. R. Baker et al. (2024) found that credit limits were decreased due to legalization, which supports this point. This mechanic provides the explanation to the aggregate debt per capita decreasing. However, the validity of the model is questioned because of the pre-trends. For online gambling legalization, I do not find a significant impact (Figure 13). The results from Table 6 support this. If anything, there is a significant impact in the fifth year, due to the size of the standard errors it is difficult to discern whether this is significant or noise.

When it comes to credit card delinquencies, the best data I have access to is the percentage of the credit card balance that is delinquent, a ratio instead of an attrition rate for credit card holders. This influences the interpretation of the model, but it is still good to investigate. When we look at the proportion of the total balance that is delinquent, we should see this narrative of restricted credit continue. As consumers gamble more, more of their balance should be delinquent. However, I find no significant effect of legalization using this measurement of delinquencies (Figure 14, 15). This may mean that a certain number of balances are becoming extremely delinquent (well over 90 days), which is the path towards default/bankruptcy which have observed in these studies.

Another interesting, yet unintuitive, finding that the economists have is that the demographics of gamblers are usually low-income and impoverished. In essence, the poor get poorer due to gambling, while other socioeconomic groups are not effected as heavily. I evaluate this claim across states using poverty rates from the University of Kentucky National Poverty Center, where I also had access to other macroeconomic variables that could be used for further analysis (University of Kentucky Center for Poverty Research 2025). Using this data, I find ample evidence in support of Hollenbeck, Larsen, and Proserpio (2024) and R. Baker et al. (2024)'s finding that the effects of gambling are most potent within the impoverished and do not cause a significant amount of people to fall into poverty. In my models, I find legalization has no effect on poverty rates (see Figure 16, 17, and Table 8). This implies that gambling is likely a symptom of impending financial collapse and bankruptcy, rather than be the driving cause.

4 Discussion and Conclusion

I use publicly available data at the state-level to run my models, which is significantly less granular than the panel data that Hollenbeck, Larsen, and Proserpio (2024) and R. Baker et al. (2024) have access to. Despite this limitation, I find sports betting does have some effect on the unemployment, namely through online legalization increasing unemployment in the long-run. I find evidence supporting the two theories behind gambling legalization, observing the substitution effect, as well as a rise in unemployment without a change in employment, possibly being new entrants to the labor market. This leads me to believe that while these two theories have some merit, there is likely another interaction that is causing the results to behave this way.

There are likely cannibalization effects due to how sports gambling can substitute other forms of casino gambling, in particular, video lottery terminals (slot machines, Keno, etc.). This has been observed in Humphreys (2021), where they found that in West Virginia, consumers would switch from video lottery terminals (VLTs) to sports betting, hurting the traditional gambling industry. Something similar could happen between retail sports betting and its online equivalent. A consumer is faced with the choice of having to go to a physical venue or gamble on their phone and may favor one over the other. This means that the employment/unemployment effects of retail and online gambling could work against each other. Many states legalize only retail due to indigenous policy, while some only legalize online sports betting. It could be that the states who permit only online betting do not have employment in mind (whether this be through direct jobs provided by the gambling industry or the substitution effect), and so an increase in the unemployment rate would be expected. This heterogeneity requires further analysis.

Moving aside from the heterogeneous effects left unanswered, the method by Sun and Abraham (2021) substantially outperforms the canonical TWFE. The staggered adoption of sports betting, looking at both retail and online legalization separately, poses a challenge that is at the forefront of current Econometrics research. This is arguably the biggest learning opportunity in the project, as I did estimate the regular TWFE model and saw how biased the results could be. The complexity of the model is needed to do the analysis properly.

In the future, I would like to do additional cross-checks to ensure that my analysis was valid. I relied on the other two papers in order to implicitly relax several assumptions, but I should also be able to do them as well. Placebo tests should have been implemented alongside my other robustness checks. Furthermore, I could have also looked at the fiscal situation of each state to see if there was indeed a timing effect in the treatment (I believe the data would have been okay for my design). I would have also liked to test other things involving debts and delinquencies that were in the Federal Reserve New York Microdata that I had access to. The papers that I read covered delinquencies spanning across several categories (credit, auto, mortgage, student), and I could have covered that as well.

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Appendices

Supplementary Figures and Tables

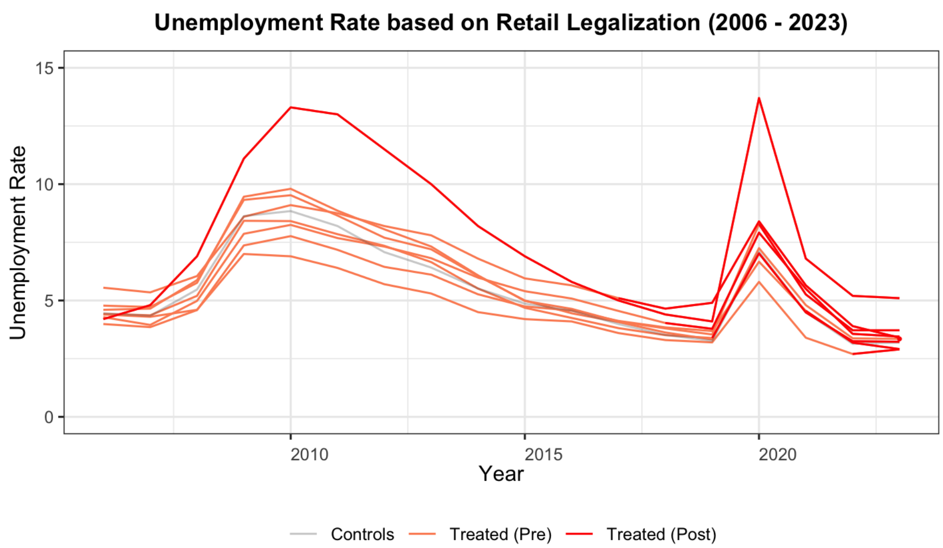

Figure 10: Unemployment of all states depending on their status. Nevada is the outlier with the highest level of unemployment regardless of economic trend.

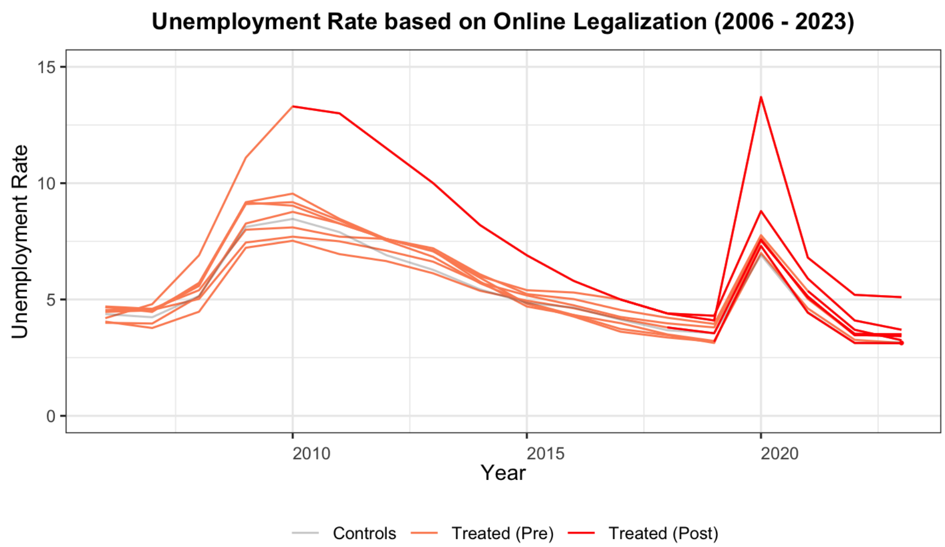

Figure 11: Unemployment of all states depending on their status. Nevada is the outlier with the highest level of unemployment regardless of economic trend.

Table 1: Descriptive Statistics of All States (2016 – 2023)

Statistics (General)	N	Mean	St. Dev.	Min	Max
Population (Thou)	392	6,652.3	7,392.7	577.7	39,522.0
Unemployment Rate	392	4.3	1.6	1.8	11.7
Bachelor's Degree (Real GDP (2017, Thou)	392	414,852.5	528,114.9	32,252.7	3,248,656.0
Employment (Thou)	392	3,117.9	3,406.6	275.6	18,627.4
Poverty Rate	392	11.2	3.1	3.7	21.5
Retail Legalized States					
Population (Thou)	280	5,889.5	5,118.1	577.7	22,904.9
Unemployment Rate (%)	280	4.4	1.6	1.8	10.1
Bachelor's Degree (% of Population)	280	33.3	5.9	20.2	47.8
Real GDP (2017, Thou)	280	364,049.5	350,960.2	36,198.2	1,791,211.0
Employment (Thou)	280	2,781.4	2,373.4	275.6	10,668.9
Poverty Rate (%)	280	11.2	3.2	3.7	21.5
Online Legalized States					
Population (Thou)	232	6,565.9	5,306.7	577.7	22,904.9
Unemployment Rate (%)	232	4.4	1.6	1.9	10.1
Bachelor's Degree (% of Population)	232	34.0	6.3	20.2	47.8
Real GDP (2017, Thou)	232	404,478.7	364,072.2	32,252.7	1,791,211.0
Employment (Thou)	232	3,093.3	2,451.0	275.6	10,668.9
Poverty Rate (%)	232	11.0	2.9	3.7	21.5

Note: Employment is not seasonally adjusted, chosen due to the nature of the sports industry. Many states that have legalized online gambling have also legalized retail gambling (and vice versa). This explains why the number of observations between the second table and third table are seemingly too high.

Table 2: Gambling on Unemployment Results

Dependent Variable:	State Unemployment Rate	
Model:	(Retail)	(Online)
Variables		
date = -7	-0.0044	0.0011
	(0.0056)	(0.0038)
date = -6	-0.0002	-0.0017
	(0.0040)	(0.0029)
date = -5	0.0049*	0.0005
	(0.0027)	(0.0026)
date = -4	0.0028	0.0004
	(0.0024)	(0.0021)
date = -3	-0.0004	-0.0004
	(0.0020)	(0.0021)
date = -2	0.0020	-0.0008
	(0.0012)	(0.0012)
date (treatment) = 0	0.0036***	0.0018
	(0.0009)	(0.0012)
date = 1	0.0013	0.0032*
	(0.0014)	(0.0018)
date = 2	0.0006	0.0022
	(0.0013)	(0.0018)
date = 3	-0.0023	0.0041*
	(0.0025)	(0.0022)
date = 4	-0.0036	0.0038*
	(0.0024)	(0.0020)
population	1.28×10^{-5}**	1.28×10^{-5}*
	(5.88×10^{-6})	(6.34×10^{-6})
rgdp_2017	-9.19×10^{-8}**	-7.9×10^{-8}*
	(3.52×10^{-8})	(3.97×10^{-8})
bachelors	0.2490***	0.2106**
	(0.0609)	(0.0907)
date = 5		0.0055**
		(0.0021)
Fit statistics		
Observations	280	239
R^2	0.92416	0.92405
Within R^2	0.29789	0.17833

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 3: Legalization on State Employment Levels

Dependent Variable:	Employment (Levels)	
Model:	(Retail)	(Online)
Variables		
date = -7	-10.60	-21.52
	(51.44)	(43.13)
date = -6	63.52	48.95
	(44.53)	(54.03)
date = -5	-20.92	0.0180
	(49.22)	(35.91)
date = -4	-3.160	-10.02
	(43.92)	(29.71)
date = -3	-17.50	-23.45
	(22.43)	(29.62)
date = -2	-11.32	-4.479
	(14.52)	(15.24)
date = 0	-36.52***	-31.99***
	(8.627)	(11.33)
date = 1	-51.43**	-20.57
	(20.18)	(17.51)
date = 2	-13.24	11.42
	(20.43)	(20.41)
date = 3	17.68	10.55
	(42.13)	(14.60)
date = 4	11.57	-24.35
	(21.56)	(17.06)
population	0.2115***	0.1792***
	(0.0481)	(0.0439)
rgdp_2017	0.0024***	0.0024***
	(0.0003)	(0.0004)
bachelors	-1,961.0**	-837.0
	(847.2)	(1,423.6)
Poverty.Rate	-2.334	-6.774
	(4.201)	(5.783)
date = 5		-11.90
		(32.83)
Fit statistics		
Observations	280	239
R^2	0.99948	0.99943
Within R^2	0.65768	0.63629

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 4: Legalization on State Unemployment (Levels)

Dependent Variable:	Unemployment (Levels)	
Model:	(Retail)	(Online)
Variables		
date = -7	-11.46	-41.90
	(56.08)	(35.25)
date = -6	6.770	-66.26*
	(24.66)	(33.65)
date = -5	39.44	-1.696
	(41.34)	(23.21)
date = -4	30.29	2.964
	(37.93)	(19.41)
date = -3	10.36	-16.98
	(15.22)	(19.89)
date = -2	17.10	-11.85
	(14.43)	(10.11)
date = 0	24.08***	21.85**
	(6.767)	(8.164)
date = 1	13.75	15.11
	(10.81)	(11.94)
date = 2	-2.309	26.28*
	(9.073)	(14.66)
date = 3	-25.92	41.31***
	(17.73)	(11.31)
date = 4	-20.60**	46.69***
	(7.973)	(7.190)
population	0.2007***	0.2200***
	(0.0469)	(0.0461)
rgdp_2017	-0.0015***	-0.0016***
	(0.0003)	(0.0003)
bachelors	1,298.2**	1,068.6
	(609.1)	(861.6)
Poverty.Rate	0.6170	3.570
	(2.522)	(3.497)
date = 5		75.21***
		(10.82)
Fit statistics		
Observations	280	239
R^2	0.93567	0.93496
Within R^2	0.36579	0.35615

Clustered (abbrev) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 5: Mean Effective Interest Rates on State Unemployment Rates

Dependent Variable:	Unemployment Rate
Model:	Mean Effective Tax Rate
Variables	
mean_eff_tax	0.0229
	(0.0229)
population	$5.86 \times 10^{-5***}$
	(1.24×10^{-5})
bachelors	0.1021
	(0.0782)
rgdp_2017	-6.62×10^{-7}
	(1×10^{-6})
Poverty.Rate	0.0007
	(0.0011)
Fixed-effects	
abbrev	Yes
date()	Yes
Fit statistics	
Observations	92
R^2	0.61090
Within R^2	0.50234

Clustered (abbrev) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 6: Legalization on Credit Card Debt Per Capita

Dependent Variable:	Credit Card Debt Per Capita	
Model:	(Retail)	(Online)
Variables		
date = -7	107.1	17.90
	(87.41)	(33.09)
date = -6	121.1**	-19.76
	(55.44)	(40.24)
date = -5	93.21	-6.922
	(71.38)	(30.07)
date = -4	94.40**	-5.512
	(45.54)	(25.92)
date = -3	42.12	0.8317
	(34.10)	(19.72)
date = -2	30.07	-1.855
	(21.11)	(11.78)
date (treatment) = 0	-21.70	2.794
	(13.74)	(11.90)
date = 1	-45.05**	4.731
	(18.43)	(18.37)
date = 2	-39.62	26.29
	(28.95)	(33.64)
date = 3	-94.67**	14.59
	(42.80)	(35.66)
date = 4	-74.74	48.14*
	(46.20)	(27.13)
date = 5		106.2**
		(43.92)
population	0.0619	0.1100
	(0.0833)	(0.0966)
rgdp_2017	0.0011***	0.0009**
	(0.0003)	(0.0004)
bachelors	-1,409.7	-870.4
	(940.1)	(968.3)
Fit statistics		
Observations	280	239
R^2	0.98965	0.99006
Within R^2	0.25544	0.26910

Clustered (State) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 7: Legalization on Percentage of Delinquent Credit Card Debt

Dependent Variable:	Percentage of Balance Delinquent	
Model:	(Retail)	(Online)
Variables		
date = -7	0.7283*	0.2041
	(0.4231)	(0.4964)
date = -6	-0.4037	-0.2302
	(0.3885)	(0.3621)
date = -5	0.0149	-0.1887
	(0.2800)	(0.2948)
date = -4	-0.1433	-0.0728
	(0.2312)	(0.2422)
date = -3	0.0734	0.0534
	(0.1383)	(0.1567)
date = -2	0.0788	0.0226
	(0.1253)	(0.1317)
date = 0	0.0682	-0.1866
	(0.1042)	(0.1317)
date = 1	0.1484	-0.1763
	(0.1369)	(0.1734)
date = 2	0.0894	-0.2827
	(0.1314)	(0.2312)
date = 3	0.3947	-0.0860
	(0.3197)	(0.3035)
date = 4	0.1833	-0.4986
	(0.3892)	(0.3587)
population	0.0002	0.0006**
	(0.0003)	(0.0003)
rgdp_2017	$-2.75 \times 10^{-6*}$	$-4.91 \times 10^{-6***}$
	(1.64×10^{-6})	(1.7×10^{-6})
bachelors	-8.037*	-8.557**
	(4.296)	(4.319)
date = 5		-0.7520
		(0.5095)
Fit statistics		
Observations	280	239
R^2	0.96143	0.96272
Within R^2	0.21176	0.25055

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 8: Legalization on the Poverty Rate

Dependent Variable:	Poverty Rate	
Model:	(Retail)	(Online)
Variables		
date = -7	-1.122	0.1039
	(1.598)	(0.8864)
date = -6	0.4605	0.5283
	(1.190)	(0.7360)
date = -5	0.4994	0.2837
	(0.8816)	(0.7692)
date = -4	-0.0990	-0.0192
	(0.6282)	(0.4374)
date = -3	-0.4026	-0.0109
	(0.4299)	(0.3228)
date = -2	-0.0617	-0.1000
	(0.2813)	(0.2645)
date = 0	-0.2891	-0.3901
	(0.2760)	(0.2821)
date = 1	-0.4688	-0.1384
	(0.3107)	(0.3445)
date = 2	0.0221	-0.2831
	(0.3765)	(0.3025)
date = 3	0.0683	-0.0271
	(0.7325)	(0.4438)
date = 4	-0.5543	0.2018
	(0.9216)	(0.4199)
population	0.0004	-7.43×10^{-5}
	(0.0007)	(0.0007)
rgdp_2017	6.61×10^{-7}	4.1×10^{-6}
	(5.67×10^{-6})	(5.37×10^{-6})
bachelors	-4.913	1.315
	(19.07)	(17.67)
unemployment	2.197	10.57
	(12.58)	(12.00)
date = 5		-0.6828
		(0.8680)
Fit statistics		
Observations	280	239
R^2	0.91976	0.89282
Within R^2	0.25848	0.19679

Clustered (abbrev) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

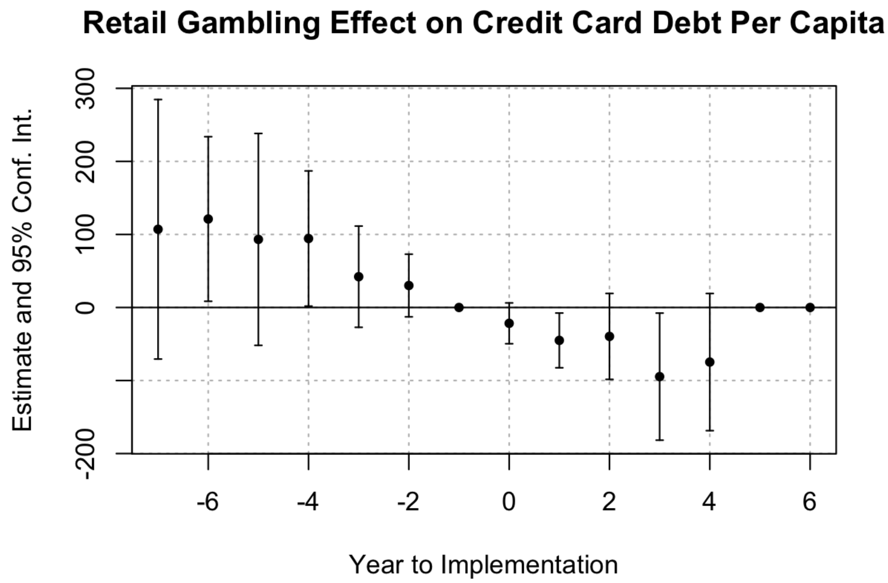

Figure 12: Retail Gambling Legalization on State Credit Card Debt (Balance Per Capita)

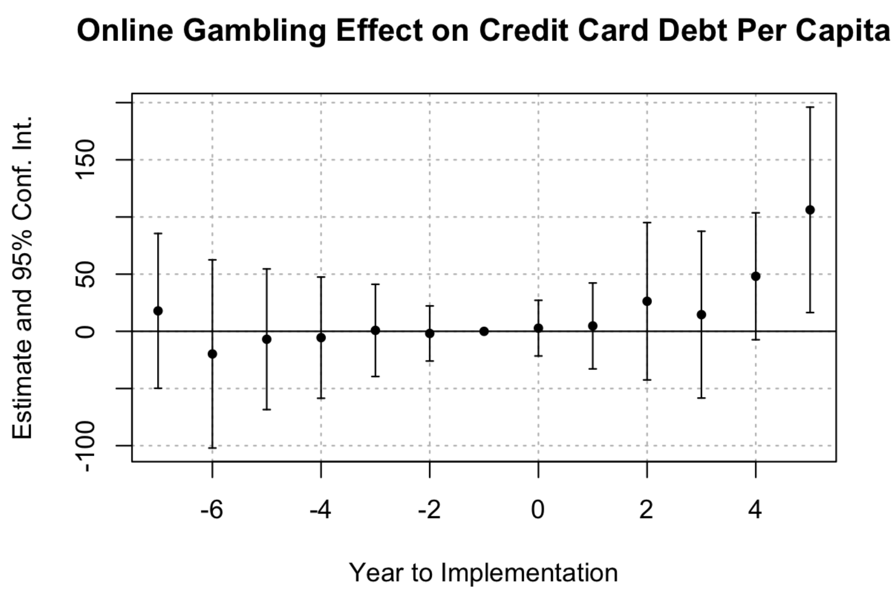

Figure 13: Online Gambling Legalization on State Credit Card Debt (Balance Per Capita)

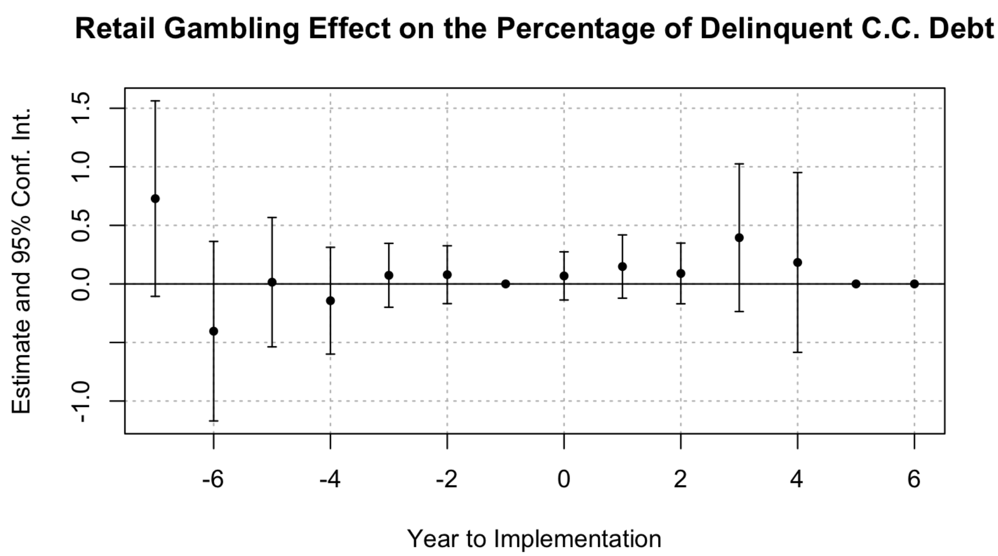

Figure 14: Retail Gambling Legalization on Credit Card Delinquencies (90+ days)

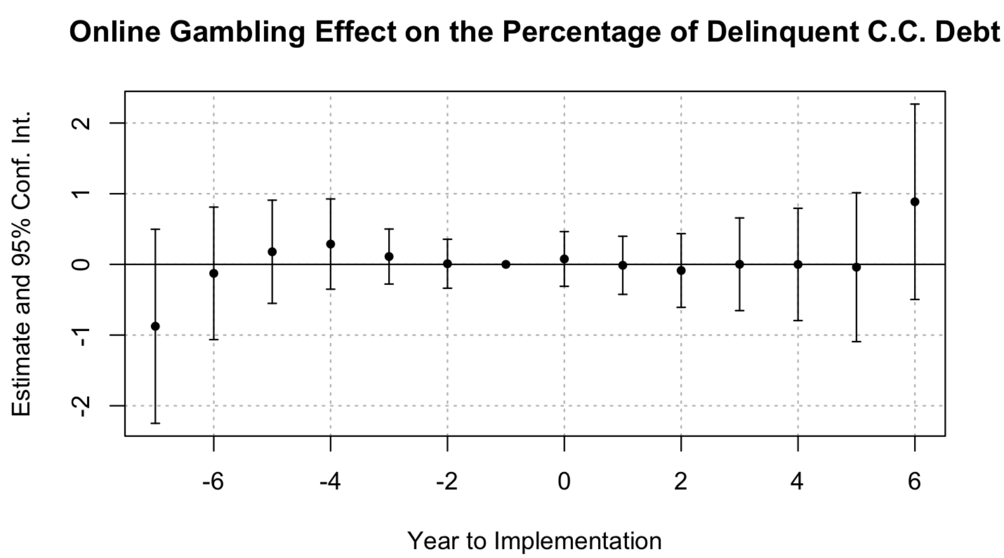

Figure 15: Online Gambling Legalization on Credit Card Delinquencies (90+ days)

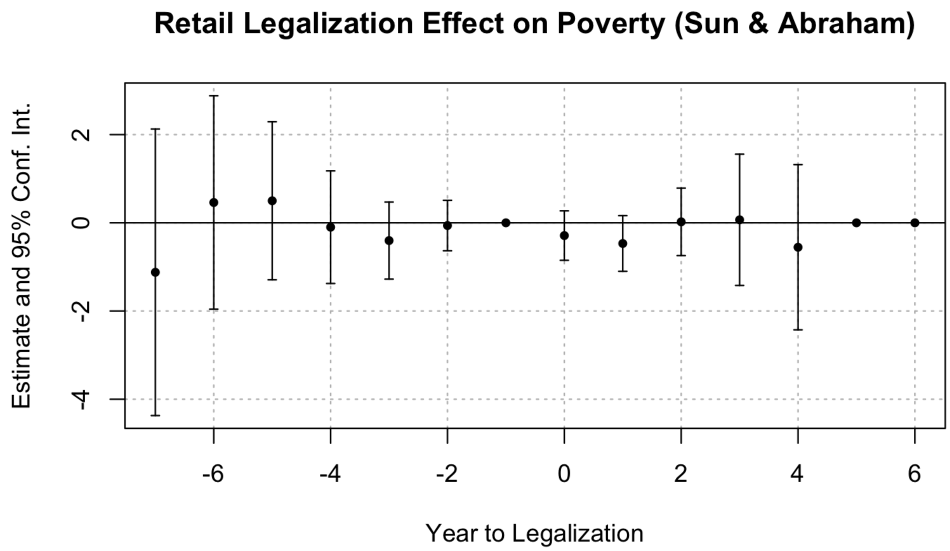

Figure 16: Retail Gambling Legalization on Poverty Rates

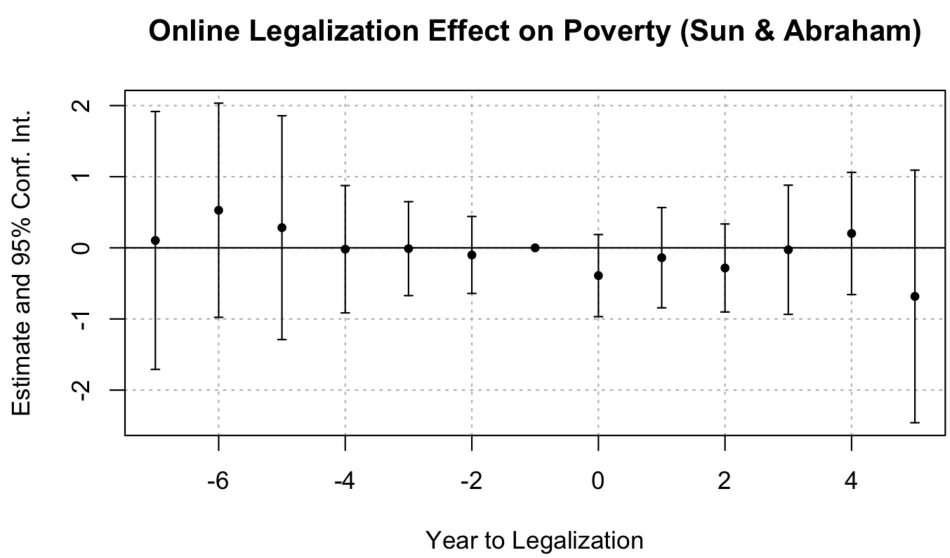

Figure 17: Online Gambling Legalization on Poverty Rates

Table 9: Average Treatment Effect on the Treated (ATT) from Legalization

	Estimate	Std. Error	t value	Pr(>|t|)
(Retail) Unemployment Rate	0.001	0.001	1.207	0.236
(Online) Unemployment Rate	0.003	0.001	2.714	0.011
(Retail) Level Employment	−25.457	16.600	−1.534	0.134
(Online) Level Employment	−13.650	10.384	−1.314	0.199
(Retail) Credit Card Debt	−44.126	25.609	−1.723	0.086
(Online) Credit Card Debt	18.218	39.157	0.465	0.642
(Retail) Delinquent C.C. Debt	0.140	0.125	1.128	0.261
(Online) Delinquent C.C. Debt	−0.244	0.182	−1.345	0.181
(Retail) Poverty Rate	−0.227	0.281	−0.810	0.424
(Online) Poverty Rate	−0.214	0.187	−1.142	0.263

The ATT is calculated from the CATT using weighted averages.

Model Specification

Sun & Abraham Models as Opposed to Static Two-Way Fixed Effects

In traditional analysis using differences in differences, the Two-Way Fixed Effects (TWFE) method is commonly used. Let us define an arbitrary static TWFE model as the regression that uses ordinary least squares (OLS).

$$y_{it} = \alpha_i + \lambda_t + \delta^{DD} D_{it} + X'_{it}\beta + \epsilon_{it} \tag{1}$$

Where we have α_i, λ_t be the unit and time fixed effects. The vector X'_{it} is the controls while β are its coefficients. The objective is to estimate δ^{DD} given that the assumptions fit for the static model. In this paper's case, the assumptions are not met. The heterogeneity of treatment effects and timing means that we opt in for some variant of

$$y_{it} = \alpha_i + \lambda_t + \sum_{l=-K}^{-2} \delta_l D^l_{it} + \sum_{l=0}^{L} \delta_l D^l_{it} + X'_{it}\beta + \epsilon_{it} \tag{2}$$

Where we take into account the time heterogeneity by specifying two sums that account for the periods before treatment as well as the periods during and after treatment. This allows the measured effects to be dynamic, since we can estimate the effect for each period.

The notation is quite verbose, but is important to show the value in Sun & Abraham's paper. When treatment occurs at the same point in time they are within the same cohort, E. When the data is binned, there is a lead/lag l that appears in the bin g (in other words, $l \in g$). The ultimate objective is to estimate the cohort average treatment effect on the treated ($CATT_{e,l}$) using the regression coefficients. We estimate both the ATT (see table 9) and the CATT in this project. Nearly all of the calculated CATT were insignificant, and were consequently excluded from the paper in place of showing the ATT, which also exhibits the same levels of insignificance (shown in Table 9).

4.0.1 Fixed Effects Model for Mean Effective Tax Rates

We have to do this for some odd reason.

$$y = \alpha_i + \gamma_t + X'_{it}\beta_{it} + \epsilon_{it} \tag{3}$$

,

Where α_i, γ_t are state and time fixed effects. The vector X_{it} includes our variable of interest, the mean effective interest rate, as well as several control variables. In this model, I controlled for the population, proportion with a Bachelor's degree, real GDP (in 2017 USD), as well as the poverty rate in each state. Results are reported in Table 5.